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(54) **INFORMATION PROCESSING APPARATUS AND INFORMATION PROCESSING METHOD**

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(57) **ABSTRACT**

An information processing apparatus includes a control unit configured to execute receiving, from a first vehicle, encrypted first data including one or more pieces of measurement data acquired by the first vehicle and one or more pieces of first dummy data that simulates the measurement data, and second data including one or more pieces of environment data each indicating a measurement environment of each piece of the measurement data and one or more pieces of second dummy data that corresponds to the first dummy data and simulates the environment data, and performing statistical processing of each piece of data included in the first data for each measurement environment indicated by each piece of data included in the second data.

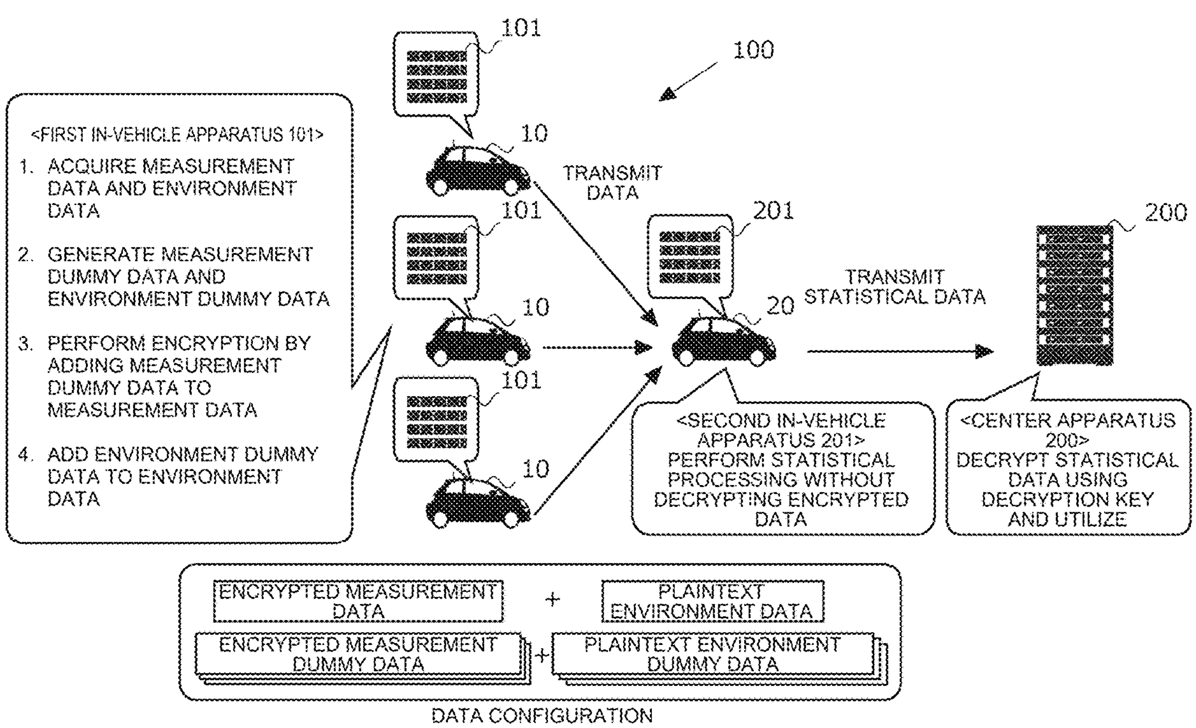
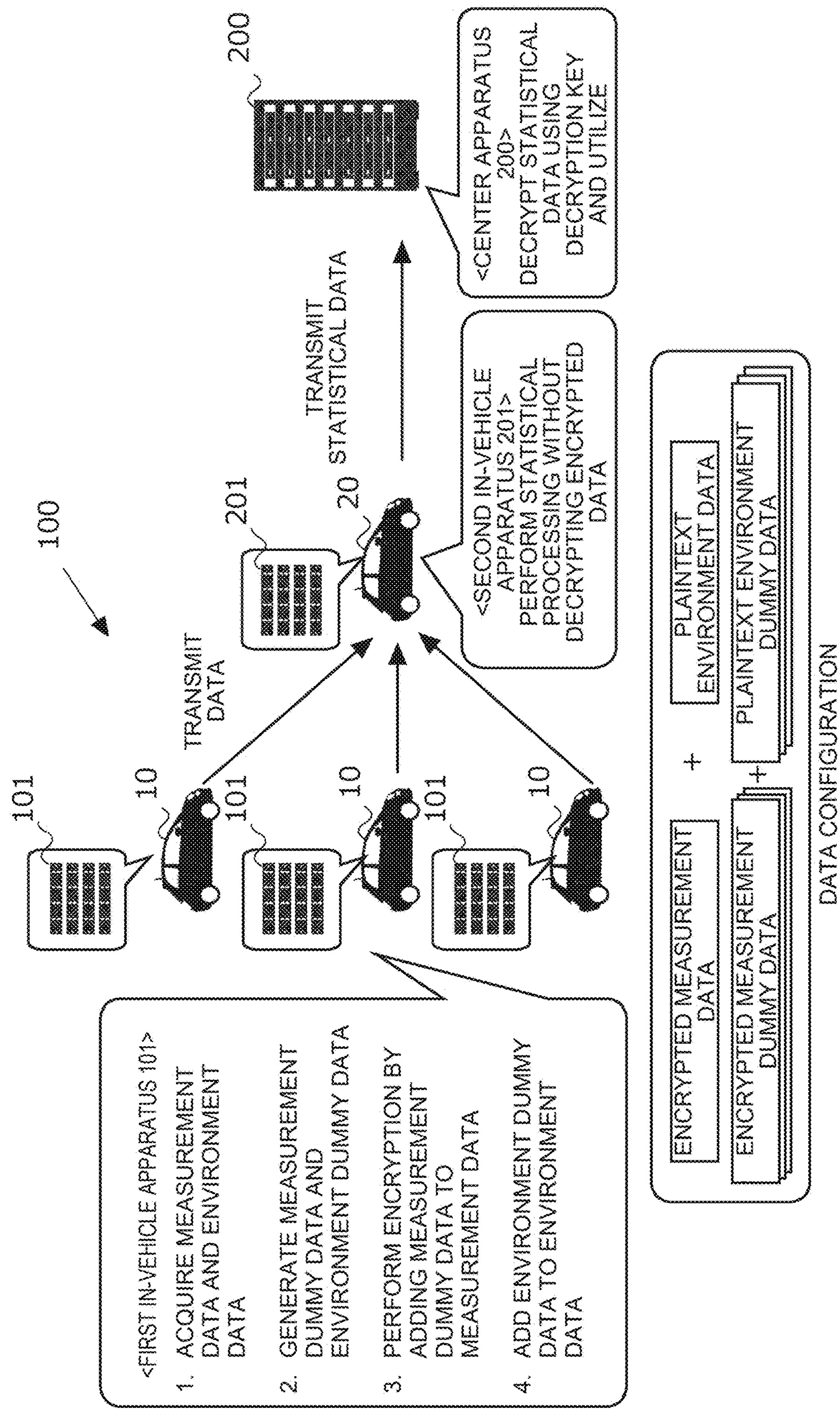


FIG. 1



100
101
10
101
10
101
10
TRANSMIT DATA
201
20
TRANSMIT STATISTICAL DATA
200
<FIRST IN-VEHICLE APPARATUS 101>
1. ACQUIRE MEASUREMENT DATA AND ENVIRONMENT DATA
2. GENERATE MEASUREMENT DUMMY DATA AND ENVIRONMENT DUMMY DATA
3. PERFORM ENCRYPTION BY ADDING MEASUREMENT DUMMY DATA TO MEASUREMENT DATA
4. ADD ENVIRONMENT DUMMY DATA TO ENVIRONMENT DATA
<SECOND IN-VEHICLE APPARATUS 201>
PERFORM STATISTICAL PROCESSING WITHOUT DECRYPTING ENCRYPTED DATA
<CENTER APPARATUS 200>
DECRYPT STATISTICAL DATA USING DECRYPTION KEY AND UTILIZE
ENCRYPTED MEASUREMENT DATA
+
PLAINTEXT ENVIRONMENT DATA
ENCRYPTED MEASUREMENT DUMMY DATA
+
PLAINTEXT ENVIRONMENT DUMMY DATA
DATA CONFIGURATION

FIG. 2

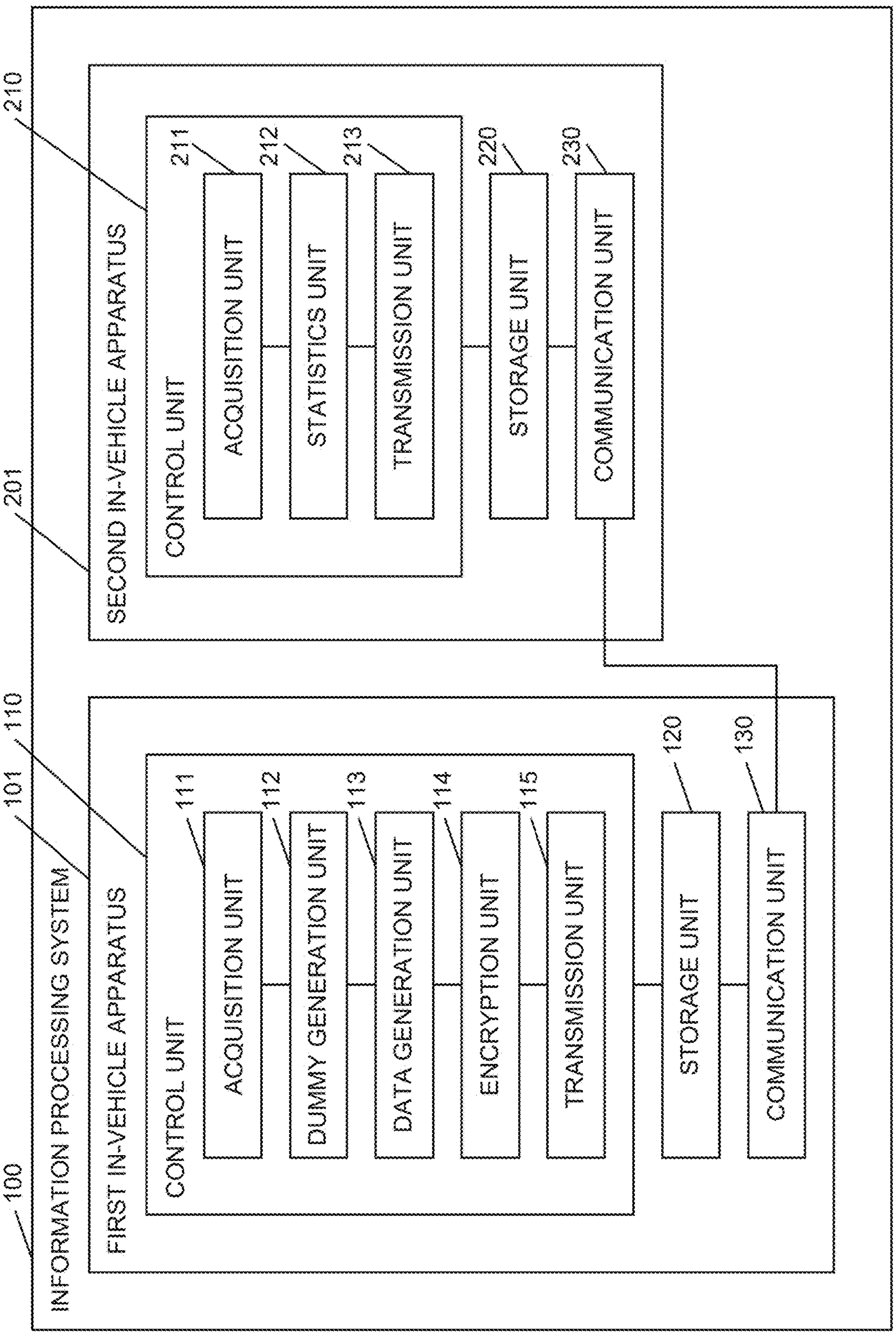

100
INFORMATION PROCESSING SYSTEM
101
FIRST IN-VEHICLE APPARATUS
110
CONTROL UNIT
111
ACQUISITION UNIT
112
DUMMY GENERATION UNIT
113
DATA GENERATION UNIT
114
ENCRYPTION UNIT
115
TRANSMISSION UNIT
120
STORAGE UNIT
130
COMMUNICATION UNIT
201
SECOND IN-VEHICLE APPARATUS
210
CONTROL UNIT
211
ACQUISITION UNIT
212
STATISTICS UNIT
213
TRANSMISSION UNIT
220
STORAGE UNIT
230
COMMUNICATION UNIT

FIG. 3

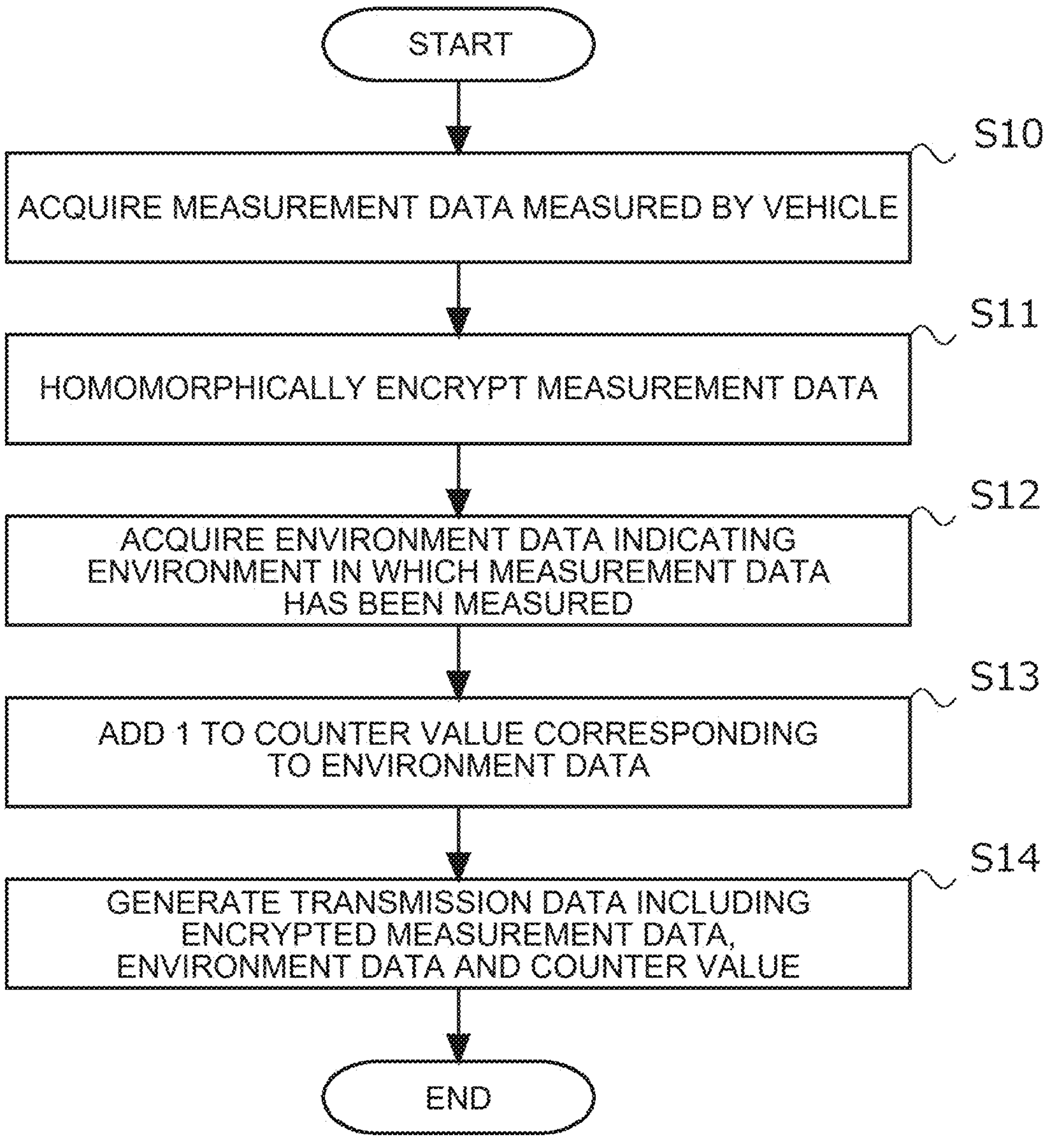

START
S10
ACQUIRE MEASUREMENT DATA MEASURED BY VEHICLE
S11
HOMOMORPHICALLY ENCRYPT MEASUREMENT DATA
S12
ACQUIRE ENVIRONMENT DATA INDICATING ENVIRONMENT IN WHICH MEASUREMENT DATA HAS BEEN MEASURED
S13
ADD 1 TO COUNTER VALUE CORRESPONDING TO ENVIRONMENT DATA
S14
GENERATE TRANSMISSION DATA INCLUDING ENCRYPTED MEASUREMENT DATA, ENVIRONMENT DATA AND COUNTER VALUE
END

FIG. 4

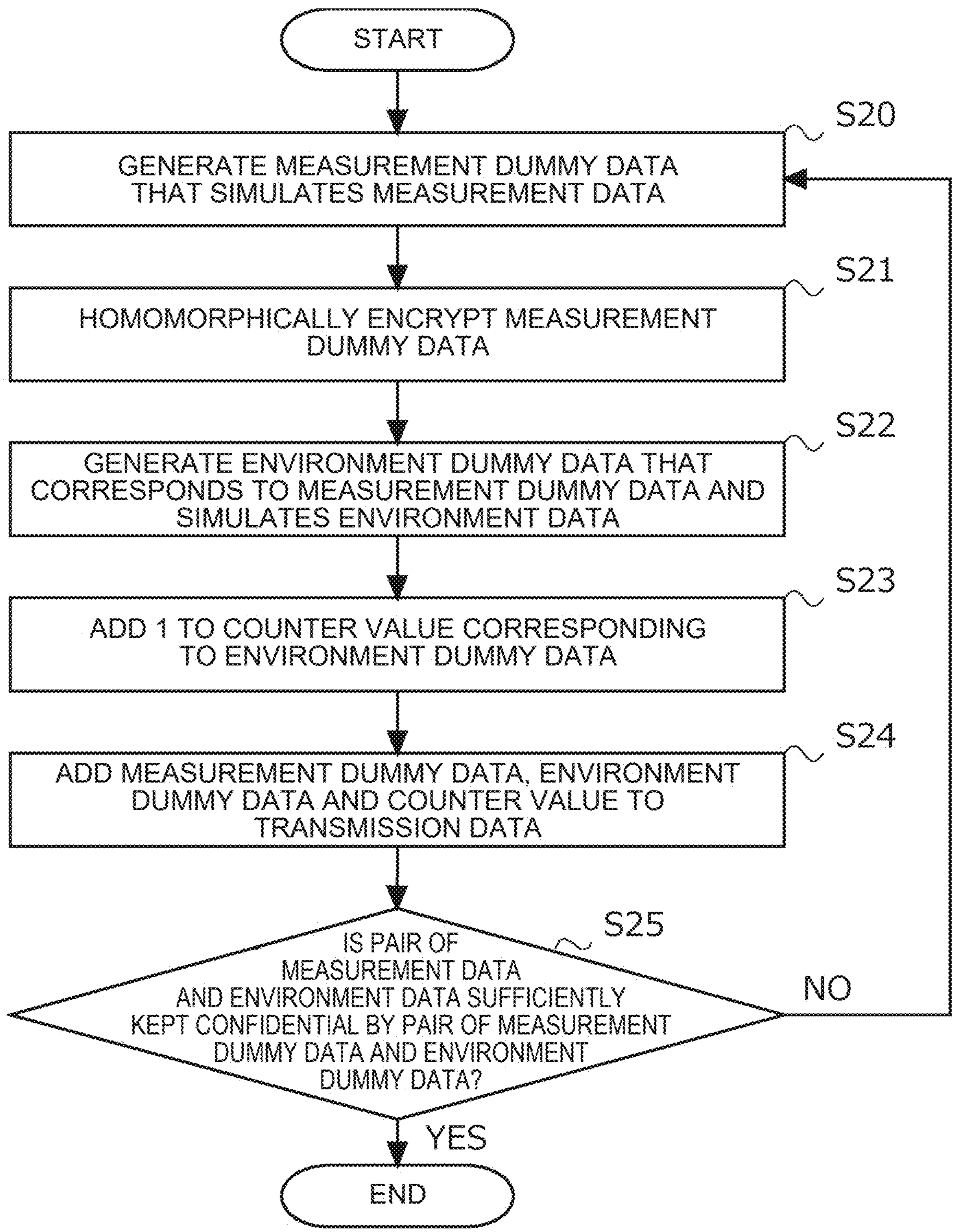
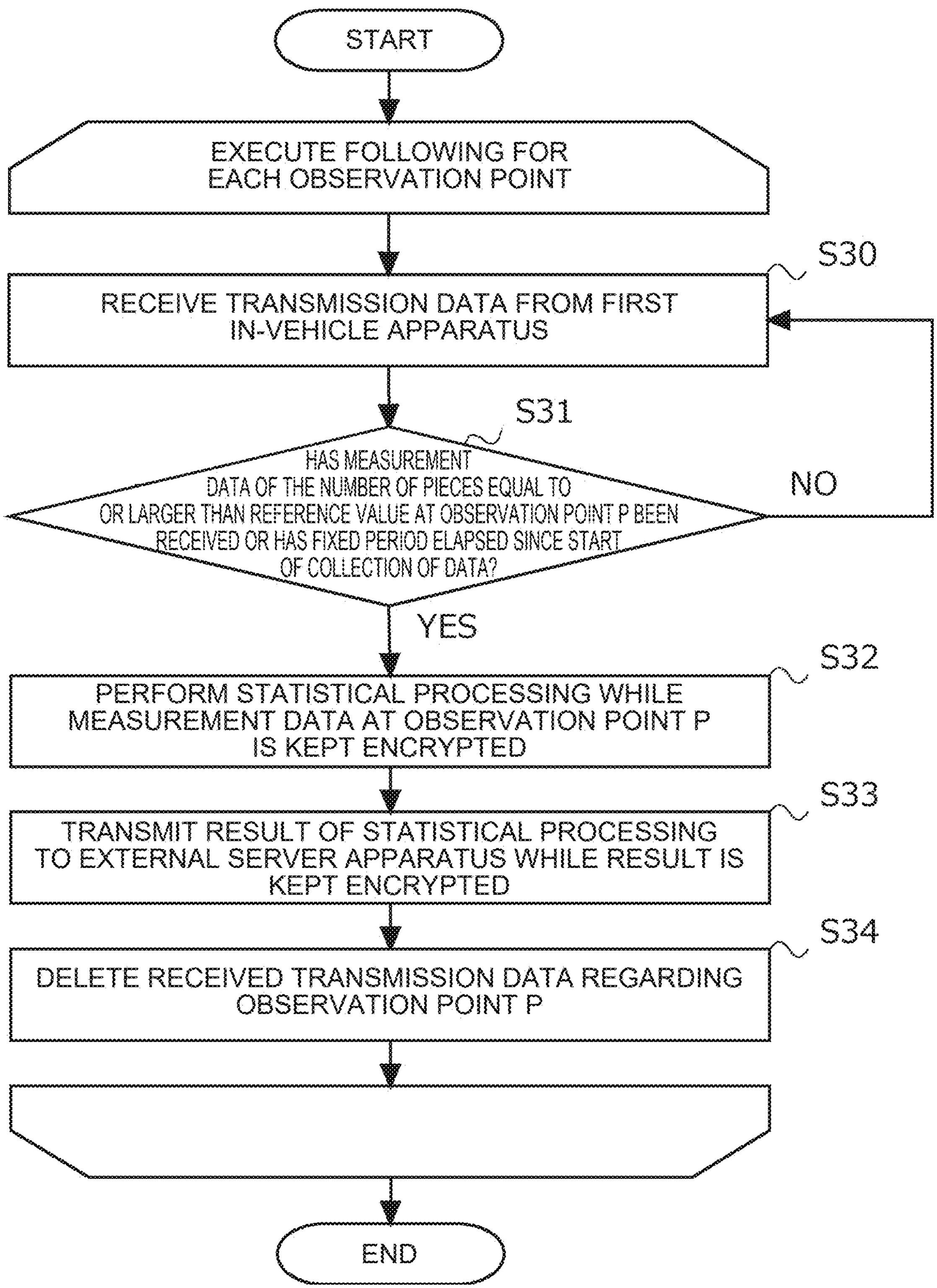

START
S20
GENERATE MEASUREMENT DUMMY DATA THAT SIMULATES MEASUREMENT DATA
S21
HOMOMORPHICALLY ENCRYPT MEASUREMENT DUMMY DATA
S22
GENERATE ENVIRONMENT DUMMY DATA THAT CORRESPONDS TO MEASUREMENT DUMMY DATA AND SIMULATES ENVIRONMENT DATA
S23
ADD 1 TO COUNTER VALUE CORRESPONDING TO ENVIRONMENT DUMMY DATA
S24
ADD MEASUREMENT DUMMY DATA, ENVIRONMENT DUMMY DATA AND COUNTER VALUE TO TRANSMISSION DATA
S25
IS PAIR OF MEASUREMENT DATA AND ENVIRONMENT DATA SUFFICIENTLY KEPT CONFIDENTIAL BY PAIR OF MEASUREMENT DUMMY DATA AND ENVIRONMENT DUMMY DATA?
NO
YES
END

FIG. 5


START
EXECUTE FOLLOWING FOR EACH OBSERVATION POINT
S30
RECEIVE TRANSMISSION DATA FROM FIRST IN-VEHICLE APPARATUS
S31
HAS MEASUREMENT DATA OF THE NUMBER OF PIECES EQUAL TO OR LARGER THAN REFERENCE VALUE AT OBSERVATION POINT P BEEN RECEIVED OR HAS FIXED PERIOD ELAPSED SINCE START OF COLLECTION OF DATA?
NO
YES
S32
PERFORM STATISTICAL PROCESSING WHILE MEASUREMENT DATA AT OBSERVATION POINT P IS KEPT ENCRYPTED
S33
TRANSMIT RESULT OF STATISTICAL PROCESSING TO EXTERNAL SERVER APPARATUS WHILE RESULT IS KEPT ENCRYPTED
S34
DELETE RECEIVED TRANSMISSION DATA REGARDING OBSERVATION POINT P
END

INFORMATION PROCESSING APPARATUS AND INFORMATION PROCESSING METHOD

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-037191 filed on Mar. 11, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to an information processing apparatus and an information processing method.

2. Description of Related Art

[0003] A technique of encrypting data and tallying the encrypted data without decrypting the encrypted data is known. Concerning this, for example, Japanese Unexamined Patent Application Publication No. 2020-119230 discloses a vehicle information collection system that acquires vehicle information, applies homomorphic encryption to the vehicle information, performs predetermined statistical processing on the encrypted vehicle information, exchanges the encrypted vehicle information with other vehicles and communicates with a center apparatus.

SUMMARY

[0004] The present disclosure is directed to tallying encrypted measurement data for each measurement environment.

[0005] One aspect of an embodiment of the present disclosure is an information processing apparatus including a control unit configured to execute receiving, from a first vehicle, encrypted first data including one or more pieces of measurement data acquired by the first vehicle and one or more pieces of first dummy data that simulates the measurement data, and second data including one or more pieces of environment data each indicating a measurement environment of each piece of the measurement data and one or more pieces of second dummy data that corresponds to the first dummy data and simulates the environment data, and performing statistical processing of each piece of data included in the first data for each measurement environment indicated by each piece of data included in the second data.

[0006] Another aspect of the embodiment of the present disclosure is an information processing apparatus mounted on a first vehicle, the information processing apparatus including a control unit configured to execute acquiring one or more pieces of measurement data generated by the first vehicle and one or more pieces of environment data each indicating a measurement environment of each piece of the measurement data, generating one or more pieces of first dummy data that simulates the measurement data acquired by the first vehicle, generating one or more pieces of second dummy data that corresponds to the first dummy data and simulates the environment data, generating first data by adding the first dummy data to one or more pieces of the measurement data, generating second data by adding the second dummy data to one or more pieces of the environment data, encrypting each piece of data included in the first data, and transmitting the first data and the second data to an information processing apparatus mounted on a second vehicle.

[0007] Yet another aspect of the embodiment of the present disclosure is an information processing method including receiving, from a first vehicle, encrypted first data including one or more pieces of measurement data acquired by the first vehicle and one or more pieces of first dummy data that simulates the measurement data, and second data including one or more pieces of environment data each indicating a measurement environment of each piece of the measurement data and one or more pieces of second dummy data that corresponds to the first dummy data and simulates the environment data, and performing statistical processing of each piece of data included in the first data for each measurement environment indicated by each piece of data included in the second data.

[0008] A still further aspect includes a program for causing a computer to execute the method described above or a computer-readable storage medium that non-transitorily stores the program.

[0009] According to the present disclosure, it is possible to tally encrypted measurement data for each measurement environment.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Features, advantages, and technical and industrial significance of exemplary embodiments of the present disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0011] FIG. **1** is a view illustrating outline of processing to be executed by an information processing system according to an embodiment;

[0012] FIG. **2** is a view for explaining components of the information processing system according to the embodiment;

[0013] FIG. **3** is a flowchart of measurement data encryption processing to be executed by a control unit of a first in-vehicle apparatus included in the information processing system according to the embodiment;

[0014] FIG. **4** is a flowchart of dummy data generation processing to be executed by the control unit of the first in-vehicle apparatus included in the information processing system according to the embodiment; and

[0015] FIG. **5** is a flowchart of processing to be executed by a control unit of a second in-vehicle apparatus included in the information processing system according to the embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

Outline

[0016] It is conceivable to utilize various kinds of data measured by a vehicle. For example, by collecting data measured by a vehicle and performing statistical processing on the collected data, it is possible to acquire data that can be utilized in traveling of the vehicle. In this case, there is a case where not a center server apparatus that originally tallies measured data, but an arbitrary vehicle receives data measured by other vehicles, performs statistical processing on the data, and transmits the result to the center server apparatus. In some embodiments, in such a case, the measured data includes information regarding privacy of a user of the vehicle that has measured the data, and thus, the vehicle that has measured the data encrypts the data and transmits the encrypted data to a vehicle that tallies the data. Even when data is encrypted, the vehicle that tallies the data can perform statistical processing in a state where the received data is kept encrypted by utilizing a specific encryption method. Typically, if data is encrypted, statistical processing cannot be performed on the data unless the data is decrypted. However, by utilizing a specific encryption method such as homomorphic encryption, statistical processing can be performed on the data without the encrypted data being decrypted.

[0017] For example, it is desired to detect whether or not an event occurs using the measured data, the event indicating a state of a road such as closure of the road, road surface freezing, sagging of the road and congestion. In this case, there is a need to acquire a statistical value of data for each measurement point (or measurement area) at which the vehicle has measured the data. However, types of calculation which is to be performed using homomorphic encryption, or the like, and which can be performed on the encrypted data are limited. Specifically, while addition, multiplication, and the like, of simple numerical values indicating a travel distance, and the like, can be performed on the encrypted data, statistics on data cannot be taken for each measurement point or each measurement area. The present disclosure provides a technique for enabling a vehicle that tallies data to tally data that is kept encrypted for each measurement point or for each measurement area and perform statistical processing.

[0018] An information processing apparatus according to one aspect of the present disclosure is an information processing apparatus including a control unit configured to execute receiving, from a first vehicle, encrypted first data including one or more pieces of measurement data acquired by the first vehicle and one or more pieces of first dummy data that simulates the measurement data, and second data including one or more pieces of environment data each indicating a measurement environment of each piece of the measurement data and one or more pieces of second dummy data that corresponds to the first dummy data and simulates the environment data, and performing statistical processing of each piece of data included in the first data for each measurement environment indicated by each piece of data included in the second data.

[0019] The measurement data is various kinds of data measured by the first vehicle using a sensor, or the like, provided in the first vehicle while the first vehicle is traveling.

[0020] The environment data is data indicating a measurement environment in which the first vehicle has acquired the measurement data. For example, the environment data may be position information indicating a position at which the first vehicle has acquired the measurement data. In this case, the environment data may include latitude and longitude information of a point at which the measurement data has been acquired.

[0021] The first dummy data is data that simulates the measurement data acquired by the first vehicle. The first dummy data includes a dummy value of data of the same type as a type of data to be detected by at least one of various kinds of sensors provided in the first vehicle.

[0022] The second dummy data is dummy data set as an environment in which the first dummy data has been measured. For example, when the environment data is position information indicating a position at which the first vehicle has acquired the measurement data, the second dummy data may include dummy latitude and longitude information corresponding to the first dummy data.

[0023] The information processing apparatus according to the present disclosure receives the first data and the second data and performs statistical processing on the encrypted first data for each measurement environment indicated by the second data that is not encrypted. In other words, the information processing apparatus according to the present disclosure can tally the encrypted measurement data for each measurement environment.

[0024] The control unit may be configured to execute the statistical processing without decrypting each piece of data included in the first data.

[0025] The information processing apparatus according to the present disclosure can protect privacy of a user of the vehicle by performing processing on data that is kept encrypted.

[0026] The control unit may be configured to perform the statistical processing on homomorphically encrypted data.

[0027] This enables the information processing apparatus according to the present disclosure to acquire measurement data that is acquired and encrypted by the first vehicle and perform statistical processing on the data that is kept encrypted.

[0028] The control unit may be configured to perform the statistical processing when the control unit determines that the first data and the second data of the number of pieces equal to or larger than a reference value have been received for one or more observation points indicated by the environment data.

[0029] This enables the information processing apparatus according to the present disclosure to secure a sufficient number of pieces of data and obtain a statistical result for the measurement data.

[0030] The control unit may be configured to calculate an average value of respective pieces of data included in the first data for each measurement environment indicated by each piece of data included in the second data in the statistical processing.

[0031] This enables the information processing apparatus according to the present disclosure to acquire tendency of data in the measurement environment on each road.

[0032] The control unit may be configured to transmit a result of the statistical processing to another information processing apparatus.

[0033] This enables the information processing apparatus according to the present disclosure to transmit the result of the statistical processing to another information processing apparatus, for example, a dedicated arithmetic apparatus or the like.

[0034] An information processing apparatus according to the present disclosure is an information processing apparatus mounted on a first vehicle, the information processing apparatus including a control unit configured to execute acquiring one or more pieces of measurement data generated by the first vehicle and one or more pieces of environment data each indicating a measurement environment of each piece of the measurement data, generating one or more pieces of first dummy data that simulates the measurement data acquired by the first vehicle, generating one or more pieces of second dummy data that corresponds to the first dummy data and simulates the environment data, generating first data by adding the first dummy data to one or more pieces of the measurement data, generating second data by adding the second dummy data to one or more pieces of the environment data, encrypting each piece of data included in the first data, and transmitting the first data and the second data to an information processing apparatus mounted on a second vehicle.

[0035] The information processing apparatus according to the present disclosure generates the first data by adding the first dummy data to the measurement data, generates the second data by adding the second dummy data to the environment data, encrypts the first data and transmits the first data and the second data.

[0036] For example, the information processing apparatus according to the present disclosure generates first data by adding to one piece of measurement data, a number of pieces of first dummy data for keeping the measurement data confidential and generates second data by adding to one piece of environment data, a number of pieces of second dummy data for keeping the environment data confidential.

[0037] This makes it impossible for the vehicle that receives the first data and the second data to estimate which is actual measurement data.

[0038] Thus, the information processing apparatus according to the present disclosure can transmit the measurement data and the environment data representing the measurement environment in which the measurement data has been acquired to other information processing apparatuses while keeping the data confidential with false data.

[0039] The control unit may be configured to encrypt each piece of data included in the first data by homomorphic encryption.

[0040] The information processing apparatus according to the present disclosure enables another information processing apparatus that has received the measurement data to perform statistical processing on the data that is kept encrypted.

[0041] The environment data may be position information of a point at which the measurement data has been acquired, and the control unit may be configured to generate position information of a point on a virtual route as the second dummy data, the virtual route being set as a traveling route of a virtual vehicle simulating the first vehicle.

[0042] The information processing apparatus according to the present disclosure can create false data that is not unnatural by generating a virtual route.

[0043] The control unit may be configured to generate position information indicating a predetermined number of positions as the second dummy data, the predetermined number of positions being selected randomly from points on peripheral roads of an observation point indicated by the environment data.

[0044] This enables the information processing apparatus according to the present disclosure to create more probable false data.

[0045] The control unit may be configured to generate a value of **0** as the first dummy data when addition calculation is utilized as the statistical processing of each piece of data included in the first data, the statistical processing being performed for each measurement environment indicated by each piece of data included in the second data, and the control unit may be configured to generate a value of **1** as the first dummy data when multiplication calculation is utilized as the statistical processing.

[0046] This enables the information processing apparatus according to the present disclosure to add to true data, a value that does not affect the statistical processing as dummy data.

[0047] The control unit may be configured to set a dummy measurement point at which the first dummy data is assumed to have been measured, determine whether or not a probability of being able to discriminate an actual measurement point is less than a predetermined threshold when the dummy measurement point and the actual measurement point indicated by the environment data are mixed, and transmit the first data and the second data to the information processing apparatus mounted on the second vehicle when the control unit determines that the probability of being able to discriminate the actual measurement point is less than the predetermined threshold.

[0048] This enables the information processing apparatus according to the present disclosure to transmit data that is reliably kept confidential by false data to other information processing apparatuses.

[0049] The information processing apparatus according to the present disclosure includes a control unit configured to execute receiving encrypted first data including one or more pieces of measurement data acquired by a first vehicle and one or more pieces of first dummy data that simulates the measurement data, second data including one or more pieces of environment data each indicating a measurement environment of each piece of the measurement data and one or more pieces of second dummy data that corresponds to the first dummy data and simulates the environment data, and a result of statistical processing performed on each piece of data included in the first data that is kept encrypted for each measurement environment indicated by each piece of data included in the second data, and decrypting the received first data and the received result.

[0050] This enables the information processing apparatus according to the present disclosure to decrypt the statistical result safely transmitted by being encrypted and utilize the decrypted statistical result.

[0051] An information processing method according to one aspect of the present disclosure includes receiving, from a first vehicle, encrypted first data including one or more pieces of measurement data acquired by the first vehicle and one or more pieces of first dummy data that simulates the measurement data, and second data including one or more pieces of environment data each indicating a measurement environment of each piece of the measurement data and one or more pieces of second dummy data that corresponds to the first dummy data and simulates the environment data, and performing statistical processing of each piece of data included in the first data for each measurement environment indicated by each piece of data included in the second data.

[0052] Performing the statistical processing may include executing the statistical processing without decrypting each piece of data included in the first data.

[0053] Performing the statistical processing may include performing the statistical processing on homomorphically encrypted data.

[0054] Performing the statistical processing may include performing the statistical processing when the first data and the second data of the number of pieces equal to or larger than a reference value are received for one or more observation points indicated by the environment data.

[0055] Performing the statistical processing may include calculating an average value of respective pieces of data included in the first data for each measurement environment indicated by each piece of data included in the second data.

[0056] The information processing method may further include transmitting a result of the statistical processing to another information processing apparatus.

[0057] An information processing method according to one aspect of the present disclosure is an information processing method to be performed by an information processing apparatus mounted on a first vehicle, the information processing method including acquiring one or more pieces of measurement data generated by the first vehicle and one or more pieces of environment data each indicating a measurement environment of each piece of the measurement data, generating one or more pieces of first dummy data that simulates the measurement data acquired by the first vehicle, generating one or more pieces of second dummy data that corresponds to the first dummy data and simulates the environment data, generating first data by adding the first dummy data to one or more pieces of the measurement data, generating second data by adding the second dummy data to one or more pieces of the environment data, encrypting each piece of data included in the first data, and transmitting the first data and the second data to an information processing apparatus mounted on a second vehicle.

[0058] A specific embodiment of the present disclosure will be described below based on the drawings. A hardware configuration, a module configuration, a functional configuration, and the like, described in each embodiment do not intend to limit a technical scope of the disclosure thereto unless otherwise described.

Embodiment

Outline of Processing of Information Processing System

[0059] Outline of processing to be performed by an information processing system according to an embodiment will be described with reference to FIG. **1**. FIG. **1** illustrates outline of processing to be executed by an information processing system **100** according to the embodiment. Here, the information processing system **100** according to the embodiment includes, for example, first in-vehicle apparatuses **101** respectively mounted on a plurality vehicles **10** that are vehicles that sense data, and a second in-vehicle apparatus **201** mounted on a vehicle **20** that is a vehicle having a function of aggregating the data. The first in-vehicle apparatus **101** and the second in-vehicle apparatus **201** may be information terminals for implementing a car navigation system or may be dedicated information terminals incorporated into the vehicle **10** and the vehicle **20**. The first in-vehicle apparatus **101** encrypts measurement data detected by a sensor, or the like, provided in the vehicle **10** and transmits the measurement data to the second in-vehicle apparatus **201** along with plaintext environment data. Then, the second in-vehicle apparatus **201** performs statistical processing on the data received from the plurality of vehicles **10** while the data is kept encrypted and transmits the received data and a result of the statistical processing, and the like, to an external server apparatus, or the like. The first in-vehicle apparatus **101** and the second in-vehicle apparatus **201** are one example of the information processing apparatus of the present disclosure. The first in-vehicle apparatus **101** is sometimes referred to as a first information processing apparatus, and the second in-vehicle apparatus **201** is sometimes referred to as a second information processing apparatus.

[0060] First, the first in-vehicle apparatus **101** of the information processing system **100** acquires measurement data and environment data via a sensor, or the like, provided in the vehicle **10**. Here, the measurement data is data acquired by the sensor. For example, the measurement data may be an image captured by an in-vehicle camera or may be speed or acceleration of the vehicle **10** detected by an acceleration sensor or data indicating whether or not an obstacle exists on a road, detected by a LiDAR, or the like, mounted on the vehicle **10**. The environment data is data indicating a measurement environment of each piece of the measurement data. Specifically, the environment data may be position information of a location where each piece of the measurement data has been measured.

[0061] Next, the first in-vehicle apparatus **101** generates dummy data respectively corresponding to the measurement data and the environment data. For example, the dummy data of the measurement data may be dummy measurement data. Specifically, the dummy data of the measurement data may be data having a value of 0 or 1. The dummy data of the measurement data is referred to as measurement dummy data. The measurement dummy data is one example of "first dummy data". Further, the dummy data of the environment data is data corresponding to the dummy data of the measurement data and indicating a dummy measurement environment set as an environment in which the dummy data has been measured. For example, the dummy data of the environment data may be dummy position information. The dummy data of the environment data is referred to as environment dummy data. The environment dummy data is one example of "second dummy data".

[0062] Subsequently, the first in-vehicle apparatus **101** adds a plurality of pieces of measurement dummy data to the measurement data and encrypts the data. For example, the first in-vehicle apparatus **101** encrypts the measurement data and the measurement dummy data using homomorphic encryption. The homomorphic encryption refers to encryption that enables calculation with encrypted text.

[0063] Then, the first in-vehicle apparatus **101** adds a plurality of pieces of environment dummy data to the environment data. The first in-vehicle apparatus **101** does not encrypt the environment data and the environment dummy data. In other words, the first in-vehicle apparatus **101** generates data by adding a plurality of pairs of encrypted measurement dummy data and plaintext environment dummy data corresponding to the encrypted measurement dummy data to a pair of encrypted measurement data and plaintext environment data corresponding to the encrypted measurement data.

[0064] Subsequently, the first in-vehicle apparatus **101** transmits a pair of the encrypted measurement data and the measurement dummy data and a pair of the environment data and the environment dummy data that are not encrypted to the second in-vehicle apparatus **201** of the vehicle **20**. Then, the second in-vehicle apparatus **201** performs statistical processing of the data that is kept encrypted without decrypting the encrypted data. For example, the second in-vehicle apparatus **201** may calculate a statistical value of the measurement data for each measurement environment indicated by the environment data corresponding to the measurement data on a one-to-one basis. Here, the statistical value may be an average value of the measurement data calculated for each measurement environment.

[0065] Then, the second in-vehicle apparatus **201** transmits statistical data to a center apparatus **200**. The center apparatus **200** decrypts the received statistical data using a decryption key and utilizes the decrypted statistical data in various kinds of calculation.

[0066] In this manner, in the present embodiment, the information processing system **100** generates data by adding a pair of encrypted measurement dummy data and plaintext environment dummy data to a pair of encrypted measurement data and plaintext environment data and performs statistical processing on the data without decrypting the data.

Configuration of Information Processing System

[0067] FIG. **2** explains components of the information processing system **100** according to the embodiment.

[0068] The information processing system **100** according to the present embodiment includes the first in-vehicle apparatus **101** and the second in-vehicle apparatus **201**. The first in-vehicle apparatus **101** includes a control unit **110**, a storage unit **120**, and a communication unit **130**. Further, the second in-vehicle apparatus **201** includes a control unit **210**, a storage unit **220**, and a communication unit **230**. In the information processing system **100**, the first in-vehicle apparatuses **101** mounted on a plurality of vehicles **10** transmit encrypted data to the second in-vehicle apparatuses **201** mounted on the vehicles **20** that are fewer than the vehicles **10**, and the second in-vehicle apparatuses **201** aggregate the received data. Note that there may be a plurality of vehicles **20**. The first in-vehicle apparatus **101** and the second in-vehicle apparatus **201** may be, for example, in-vehicle terminals that implement a car navigation system, or the like, or may be dedicated in-vehicle terminals for executing processing described in the present embodiment.

[0069] The vehicle **20** is a vehicle that takes statistics of the data generated by the vehicle **10**. The vehicle **20** may be selected by a management apparatus that manages the information processing system **100**. For example, when statistics are taken for each of a plurality of areas, the management apparatus may select the vehicle **20** from a plurality of vehicles traveling in a predetermined area.

[0070] First, components of the first in-vehicle apparatus **101** will be described.

[0071] The control unit **110** is implemented with a processor such as a central processing unit (CPU) or a graphics processing unit (GPU) and a memory. The control unit **110** includes an acquisition unit **111**, a dummy generation unit **112**, a data generation unit **113**, an encryption unit **114**, and a transmission unit **115** as functional modules. These functional modules may be implemented by programs being executed by the control unit **110**. Note that the control unit **110** is sometimes referred to as a first control unit.

[0072] The acquisition unit **111** acquires measurement data acquired by a sensor, or the like, provided in the vehicle **10**, and environment data indicating a measurement environment of the measurement data. The acquisition unit **111** may perform communication with an electronic control unit (ECU) that controls the sensor, or the like, provided in the vehicle **10** via the communication unit **130** to acquire the measurement data and the environment data.

[0073] The dummy generation unit **112** generates measurement dummy data that is dummy data simulating the measurement data and environment dummy data that is dummy data corresponding to the measurement dummy data and simulating the environment data. The dummy generation unit **112** can set a value that does not affect statistical calculation of the measurement data as the measurement dummy data. Specifically, the dummy generation unit **112** can set a value of 0 as the measurement dummy data when addition calculation is utilized as the statistical processing of the measurement data and can set a value of 1 as the measurement dummy data when multiplication calculation is utilized as the statistical processing of the measurement data.

[0074] The data generation unit **113** generates first data by adding one or more pieces of the measurement dummy data to one or more pieces of the measurement data. Further, the data generation unit **113** generates second data by adding one or more pieces of the environment dummy data corresponding to the measurement dummy data to one or more pieces of the environment data corresponding to the measurement data. In other words, the data generation unit **113** generates data by adding a pair of one or more pieces of measurement dummy data and environment dummy data to a pair of one or more pieces of the measurement data and the environment data.

[0075] The encryption unit **114** encrypts each piece of data included in the first data. In other words, the encryption unit **114** encrypts the measurement data and the measurement dummy data. The encryption unit **114** may encrypt each piece of data included in the first data using homomorphic encryption.

[0076] The transmission unit **115** transmits a pair of one or more pieces of the encrypted first data and plaintext second data to the second in-vehicle apparatus **201**. The transmission unit **115** performs wireless communication with the communication unit **230** of the second in-vehicle apparatus **201** via the communication unit **130**. The transmission unit **115** may transmit the data to the second in-vehicle apparatus **201** through communication via a wide area network or may transmit the data to the second in-vehicle apparatus **201** through vehicle-to-vehicle communication.

[0077] The storage unit **120** is a main storage device such as a random access memory (RAM) or a read only memory (ROM), or an auxiliary storage device such as an erasable programmable ROM (EPROM), a hard disk drive and a removable medium. In the auxiliary storage device, an operating system (OS), various kinds of programs, various kinds of tables, and the like, are stored, and respective functions that match predetermined purposes of the respective units of the control unit **110** can be implemented by the programs stored therein being executed. However, some or all of the functions may be implemented by a hardware circuit such as an application specific integrated circuit (ASIC) or a field programmable gate array (FPGA).

[0078] The storage unit **120** stores data, and the like, to be used or generated in processing to be performed by the control unit **110**. The storage unit **120** may store programs, and the like, to be used for encrypting data.

[0079] The communication unit **130** is constituted with a communication circuit that performs wireless communication. The communication unit **130** may be, for example, a communication circuit that performs wireless communication using 4th generation (4G) or may be a communication circuit that performs wireless communication using 5th generation (5G). The communication unit **130** may be a communication circuit that performs wireless communication using long term evolution (LTE) or may be a communication circuit that performs communication using low power wide area (LPWA). The communication unit **130** may be a communication circuit that performs wireless communication using Wi-Fi (registered trademark).

[0080] Components of the second in-vehicle apparatus **201** will be described next.

[0081] The control unit **210** is implemented with a processor such as a CPU or a GPU and a memory. The control unit **210** includes an acquisition unit **211**, a statistics unit **212** and a transmission unit **213** as functional modules. These functional modules may be implemented by programs being executed by the control unit **210**. Note that the control unit **210** is sometimes referred to as a second control unit.

[0082] The acquisition unit **211** acquires one or more pieces of encrypted first data and plaintext second data from the first in-vehicle apparatus **101**. The acquisition unit **211** may receive one or more pieces of encrypted first data and plaintext second data from the first in-vehicle apparatus **101** via the communication unit **230**.

[0083] The statistics unit **212** performs statistical processing of each piece of data included in the first data for each measurement environment indicated by each piece of data included in the second data. The statistics unit **212** performs statistical processing on the first data that is kept encrypted based on each piece of data included in the plaintext second data. For example, the statistics unit **212** may calculate an average value of the measurement data for each measurement point (or measurement area) of each piece of the measurement data indicated by the second data without decrypting the measurement data.

[0084] The transmission unit **213** transmits statistical data calculated by the statistics unit **212** to an external server apparatus, or the like. Here, the external server apparatus, or the like, is a center apparatus, or the like, that performs calculation to utilize the measurement data measured by the vehicle **10** in various kinds of applications. The transmission unit **213** transmits the statistical data to the external server apparatus, or the like, via a wide area network such as the Internet via the communication unit **230**.

[0085] The storage unit **220** is a main storage device such as a RAM or a ROM or an auxiliary storage device such as an EPROM, a hard disk drive and a removable medium.

[0086] In the auxiliary storage device, an operating system (OS), various kinds of programs, various kinds of tables, and the like, are stored, and respective functions that match predetermined purposes of the respective units of the control unit **210** can be implemented by the programs stored therein being executed. However, some or all of the functions may be implemented by a hardware circuit such as an ASIC and an FPGA.

[0087] The storage unit **220** stores data, and the like, to be used or generated in processing to be performed by the control unit **210**. The storage unit **220** may store programs, and the like, to be used in statistical processing of the first data.

[0088] The communication unit **230** is constituted with a communication circuit that performs wireless communication. The communication unit **230** may be, for example, a communication circuit that performs wireless communication using 4G or may be a communication circuit that performs wireless communication using 5G. Alternatively, the communication unit **230** may be a communication circuit that performs wireless communication using LTE or may be a communication circuit that performs communication using LPWA. Alternatively, the communication unit **230** may be a communication circuit that performs wireless communication using Wi-Fi (registered trademark).

Operation of First In-Vehicle Apparatus

[0089] Specific content of processing to be performed by the first in-vehicle apparatus **101** included in the information processing system **100** will be described next. First, processing of the first in-vehicle apparatus **101** encrypting data measured by the vehicle **10** will be described. FIG. **3** is a flowchart of measurement data encryption processing to be executed by the control unit **110** of the first in-vehicle apparatus **101** included in the information processing system **100** according to the embodiment.

[0090] The first in-vehicle apparatus **101** included in the information processing system **100** starts processing described in FIG. **3** when the vehicle **10** on which the first in-vehicle apparatus **101** is mounted starts traveling, and periodically repeats the processing at regular intervals. Note that the first in-vehicle apparatus **101** may start processing in step **S10** when the first in-vehicle apparatus **101** accepts some kind of input from a user of the vehicle **10**.

[0091] In step S**10**, the acquisition unit **111** acquires measurement data via a sensor provided in the vehicle **10**. Here, the measurement data may be specifically an image captured by an in-vehicle camera. Alternatively, the measurement data may be speed or acceleration of the vehicle **10** detected by an acceleration sensor or data indicating whether or not an obstacle exists on a road, detected by a LiDAR, or the like, mounted on the vehicle **10**.

[0092] In step S**11**, the encryption unit **114** homomorphically encrypts the measurement data acquired by the acquisition unit **111** in step **S10**. Here, as the homomorphic encryption, the Okamoto-Uchiyama cryptosystem, the Paillier cryptosystem, the ElGamal cryptosystem, RSA encryption, the Boneh-Goh-Nissim cryptosystem, lattice cryptography, fully homomorphic encryption based on lattice cryptography by Gentry, or the like, may be used.

[0093] In step S**12**, the acquisition unit **111** acquires environment data indicating an environment in which the measurement data acquired in step **S10** has been measured. One piece of environment data corresponds to one piece of measurement data. The environment data may be, for example, position information indicating a position at which each piece of the measurement data has been measured. Specifically, the environment data may be latitude and longitude information (which may include altitude information) of the position at which the measurement data has been measured. The environment data is not limited to the position information indicating the position at which the measurement data has been measured and may be any data as long as the data indicates the environment in which the measurement data has been measured.

[0094] For example, the environment data may be information indicating hours in which the measurement data has been measured, an ID of a road link of a road on which the measurement data has been acquired, or a lane of the road on which the measurement data has been acquired. Further, for example, the environment data may be information indicating a traveling direction of the vehicle that has acquired the measurement data when the measurement data has been acquired, vehicle speed of the vehicle that has acquired the measurement data when the measurement data has been acquired, a vehicle model or a vehicle type of the vehicle that has acquired the measurement data, or the like.

[0095] In step S**13**, the data generation unit **113** assigns 1 to a counter value corresponding to the environment data. The counter value is data to be transmitted to the second in-vehicle apparatus **201** by the first in-vehicle apparatus **101** along with the measurement data and the corresponding environment data. When a sensor mounted on the vehicle **10** performs sensing once in a certain measurement environment, 1 is added to the counter value. In other words, the counter value indicates the number of pieces of the acquired measurement data in the measurement environment indicated by the environment data. The second in-vehicle apparatus **201** can recognize how many pieces of measurement data corresponding to a specific measurement environment are collected by summing the counter values included in the collected transmission data.

[0096] In step S**14**, the data generation unit **113** generates transmission data including the measurement data encrypted in step S**11**, the environment data corresponding to the measurement data, and the counter value to which **1** is assigned in step S**13**. The data generation unit **113** may make a pair of the encrypted measurement data and plaintext environment data corresponding to the measurement data to make the transmission data or may make the transmission data including a plaintext counter value separately from the pair of the measurement data and the environment data.

[0097] Processing of the first in-vehicle apparatus **101** generating dummy data for keeping the data measured by the vehicle **10** confidential will be described next. FIG. **4** is a flowchart of dummy data generation processing to be executed by the control unit **110** of the first in-vehicle apparatus **101** included in the information processing system **100** according to the embodiment.

[0098] The first in-vehicle apparatus **101** included in the information processing system **100** starts processing described in FIG. **4** when the vehicle **10** on which the first in-vehicle apparatus **101** is mounted starts traveling, and periodically repeats the processing at regular intervals. Note that the first in-vehicle apparatus **101** may start processing in step S**20** when the first in-vehicle apparatus **101** accepts some kind of input from the user of the vehicle **10**. The processing described in FIG. **4** may be performed in parallel to the processing described in FIG. **3** or may be performed after the processing described in FIG. **3**. When the processing described in FIG. **3** and the processing described in FIG. **4** are repeated, the processing described in FIG. **3** and the processing described in FIG. **4** may be alternately performed.

[0099] In step S**20**, the dummy generation unit **112** generates measurement dummy data that is dummy data simulating the measurement data. The measurement dummy data has a value that does not affect a calculation result in calculation of performing statistical processing on the measurement data. For example, when the statistical processing on the measurement data includes addition calculation, the measurement dummy data may have a value of zero. Further, when the statistical processing on the measurement data includes multiplication calculation, the measurement dummy data may have a value of one.

[0100] In step S**21**, the encryption unit **114** homomorphically encrypts the measurement dummy data. The encryption unit **114** encrypts the measurement dummy data using the same method as the method in step S**11**.

[0101] In step S**22**, the dummy generation unit **112** generates environment dummy data that corresponds to the measurement dummy data and simulates the environment data. Specifically, the dummy generation unit **112** generates the environment dummy data indicating a value indicating an environment set as an environment in which the measurement dummy data has been measured. For example, the dummy generation unit **112** generates the environment dummy data indicating position information (for example, latitude and longitude information (which may include altitude information)) indicating a position set as a position at which the measurement dummy data has been measured. Specifically, as the environment dummy data, position information indicating a predetermined number of positions randomly selected from points on peripheral roads of an actual observation point indicated by true environment data may be employed. Alternatively, the dummy generation unit **112** sets a predetermined number of dummy vehicles and generates dummy trajectories on which the dummy vehicles have traveled. Then, a predetermined number of points may be extracted on the dummy trajectories, and position information indicating the positions may be employed as the environment dummy data.

[0102] In step S**23**, the data generation unit **113** assigns 1 to a counter value corresponding to the environment dummy data. In other words, the counter value indicates that (it is set as) one piece of measurement dummy data has been acquired in the measurement environment indicated by the environment dummy data.

[0103] In step S**24**, the data generation unit **113** adds the measurement dummy data encrypted in step S**21**, the environment dummy data corresponding to the measurement dummy data and the counter value to which 1 is assigned in step S**23** to the transmission data. The data generation unit **113** may make a pair of the encrypted measurement dummy data and the plaintext environment dummy data corresponding to the measurement dummy data and add the pair to the transmission data. Here, the transmission data is transmission data generated in the processing (step S**14**) in FIG. **3**.

[0104] In step S**25**, the data generation unit **113** determines whether or not the pair of the measurement data and the environment data is sufficiently kept confidential by the pair of the measurement dummy data and the environment dummy data in the transmission data generated in step S**24**.

[0105] For example, the data generation unit **113** may determine whether or not the number of pairs of the measurement dummy data and the environment dummy data is larger than the number of pairs of the measurement data and the environment data by a given number in the transmission data. Alternatively, the data generation unit **113** may determine whether or not a ratio of the number of pairs of the measurement dummy data and the environment dummy data with respect to the number of pairs of the measurement data and the environment data is equal to or larger than a given ratio in the transmission data. Alternatively, the data generation unit **113** may determine whether or not a probability of being able to discriminate the actual measurement point is less than a predetermined threshold when the dummy measurement point employed as the environment dummy data and the actual measurement point indicated by the environment data are mixed.

[0106] When the data generation unit **113** determines that the pair of the measurement data and the environment data is sufficiently kept confidential by the pair of the measurement dummy data and the environment dummy data in the transmission data, a positive determination result is obtained in the present step.

[0107] When the positive determination result is obtained in the present step, the processing ends.

[0108] When a negative determination result is obtained in the present step, the processing transitions to step S**20**.

[0109] If both the processing in FIG. **3** and the processing in FIG. **4** end, the transmission unit **115** transmits the transmission data, that is, data including one or more pairs of the measurement data and the environment data and one or more pairs of the measurement dummy data and the environment dummy data to the second in-vehicle apparatus **201**. Further, when the processing described in FIG. **3** and the processing described in FIG. **4** are repeatedly performed, the processing may be performed as follows. In this case, at a time point at which both the processing in step S**14** and processing in step S**25** that is performed the same number of times as the processing in step S**14** are completed, the data including one or more pairs of the measurement data and the environment data and one or more pairs of the measurement dummy data and the environment dummy data may be transmitted to the second in-vehicle apparatus **201**. The above processing may be repeatedly performed at the above-described timing.

Operation of Second In-Vehicle Apparatus

[0110] Specific content of processing to be performed by the second in-vehicle apparatus **201** included in the information processing system **100** will be described next. Specifically, operation of the second in-vehicle apparatus **201** performing statistical processing on the data received by the vehicle **20** will be described. FIG. **5** is a flowchart of processing to be executed by the control unit **210** of the second in-vehicle apparatus **201** included in the information processing system **100** according to the embodiment. The control unit **210** repeats processing from step S**30** to step S**34** for each observation point. Here, the observation point may be a local point or a given range (observation area). "Each observation point" may be a point or an area determined in advance. For example, when a target region includes a plurality of regions divided by mesh, the processing from step S**30** to S**34** may be repeated for each of the plurality of regions.

[0111] In step S**30**, the acquisition unit **211** receives the transmission data from the first in-vehicle apparatus **101** mounted on the vehicle **10**. The acquisition unit **211** receives the transmission data generated by the first in-vehicle apparatus **101** in step S**14** and step S**25**. In other words, the acquisition unit **211** receives data including one or more pairs of the measurement data and the environment data and one or more pairs of the measurement dummy data and the environment dummy data, the data being transmitted to the second in-vehicle apparatus **201** by the first in-vehicle apparatus **101**.

[0112] In step S**31**, the statistics unit **212** determines whether data of the number of pieces equal to or larger than a reference value has been received or a fixed period has elapsed since start of collection of the measurement data for the measurement data acquired at the target observation point. The environment data is plaintext, so that the statistics unit **212** can determine the observation point corresponding to each of a plurality of pieces of the measurement data. For example, when the target region includes a plurality of regions divided by mesh, whether a predetermined number of pieces of measurement data have been acquired may be determined for each of the plurality of regions.

[0113] When the statistics unit **212** determines that data of the number of pieces equal to or larger than the reference value has been received or a fixed period has elapsed since start of collection of the measurement data for the measurement data acquired at the target observation point, a positive determination result is obtained in the present step.

[0114] When the positive determination result is obtained in the present step, the processing transitions to step S**32**.

[0115] When a negative determination result is obtained in the present step, the processing transitions to step S**30**.

[0116] When the processing transitions to step S**32**, the statistics unit **212** performs statistical processing on a given number of pieces of measurement data acquired at the observation point P while the measurement data is kept encrypted. For example, the statistics unit **212** may perform processing of calculating an average value for the measurement data that is acquired at the observation point P and is kept encrypted.

[0117] Then, in step S**33**, the transmission unit **213** transmits a result of the statistical processing calculated in step S**32**, which is data that is kept encrypted, to an external server apparatus. The transmission unit **213** transmits the encrypted result of the statistical processing to an external server apparatus such as a center apparatus having a function of receiving data from the vehicle **10** and the vehicle **20** and utilizing the received data in various kinds of calculation. The result of the statistical processing may be transmitted along with data indicating the corresponding observation point (point or area). Further, the transmission unit **213** may transmit the data itself received from the first in-vehicle apparatus **101** to the external server apparatus at the same time as well as the result of the statistical processing. The received data relating to the target observation point is stored in the storage unit **220** by the control unit **210**. In step S**34**, the transmitted data relating to the target observation point may be deleted by the control unit **210**.

[0118] The center apparatus, or the like, that has received the result of the statistical processing from the second in-vehicle apparatus **201** decrypts the encrypted result of the statistical processing using a decryption key. Then, the center apparatus, or the like, can utilize the decrypted statistical value in various kinds of calculation.

[0119] In the present embodiment, the information processing system **100** can keep the true measurement data and environment data confidential by the dummy data that simulates the measurement data and the environment data measured by the vehicle **10** and perform statistical processing of the measurement data while keeping the data encrypted. In other words, the information processing system **100** can transmit measurement data regarding privacy of the user of the vehicle **10** to the vehicle **20** in a state where the measurement data is kept confidential, and the vehicle **20** can perform statistical processing on the measurement data that is kept confidential.

Modification 1

[0120] In the above embodiment, all the first in-vehicle apparatuses **101** transmit measurement data measured in various measurement environments to the second in-vehicle apparatus **201**. However, only the first in-vehicle apparatus **101** of the vehicle **10** that satisfies a specific measurement position or a measurement condition designated from the center apparatus, or the like, may transmit only the measurement data that satisfies the condition to the second in-vehicle apparatus **201**. In other words, when the center apparatus designates the measurement position, the measurement condition, and the like, only the first in-vehicle apparatus **101** mounted on the vehicle **10** that can satisfy the condition encrypts the measurement data and transmits the encrypted measurement data to the second in-vehicle apparatus **201**. In this event, the first in-vehicle apparatus **101** may transmit only the measurement data to the second in-vehicle apparatus **201** without adding the environment data to the transmission data. This is because the measurement position, the measurement condition, and the like, are limited in advance, and thus, statistical processing can be performed as is without classifying the measurement data for each measurement position or measurement environment.

[0121] However, if only the first in-vehicle apparatus **101** of a specific vehicle **10** that satisfies the measurement condition transmits the transmission data to the second in-vehicle apparatus **201**, there is a possibility that history of the position information of the vehicle **10**, history of other states, or the like, may be acquired by the vehicle **20**. Thus, also the first in-vehicle apparatus **101** mounted on the vehicle **10** that does not satisfy the designated measurement condition also transmits the encrypted measurement dummy data to the second in-vehicle apparatus **201** at a constant rate. Then, the second in-vehicle apparatus **201** performs statistical processing while the measurement data received from the specific vehicle **10** that satisfies the measurement condition and the measurement dummy data received from the vehicle **10** that does not satisfy the measurement condition are kept encrypted. As such, it is possible to acquire a statistical value of the measurement data acquired in a predetermined measurement environment while preventing leakage of information regarding privacy of the specific vehicle **10** that satisfies the measurement condition.

Other Modifications

[0122] The above embodiment is merely one example, and the present disclosure can be implemented while change is made as appropriate within a range not deviating from the gist.

[0123] The present disclosure can be implemented by a computer program implementing the functions described in the above embodiment being supplied to a computer and one or more processors of the computer reading and executing the program. Such a computer program may be provided to the computer using a non-transitory computer-readable storage medium that can be connected to a system bus of the computer or may be provided to the computer via a network. The non-transitory computer-readable storage medium includes, for example, an arbitrary type of disk such as a magnetic disk (such as a floppy (registered trademark) disk and a hard disk drive (HDD)) and an optical disk (such as a CD-ROM, a DVD disk and a blue-ray disk), a read only memory (ROM), a random access memory (RAM), an EPROM, an EEPROM, a magnetic card, a flash memory, an optical card, and an arbitrary type of medium appropriate for storing an electronic command.

What is claimed is:

1. An information processing apparatus comprising a control unit configured to execute:

receiving, from a first vehicle, encrypted first data including one or more pieces of measurement data acquired by the first vehicle and one or more pieces of first dummy data that simulates the measurement data, and second data including one or more pieces of environment data each indicating a measurement environment of each piece of the measurement data and one or more pieces of second dummy data that corresponds to the first dummy data and simulates the environment data; and

performing statistical processing of each piece of data included in the first data for each measurement environment indicated by each piece of data included in the second data.

2. The information processing apparatus according to claim **1**, wherein the control unit is configured to execute the statistical processing without decrypting each piece of data included in the first data.

3. The information processing apparatus according to claim **2**, wherein the control unit is configured to perform the statistical processing on homomorphically encrypted data.

4. The information processing apparatus according to claim **1**, wherein the control unit is configured to perform the statistical processing when the control unit determines that the first data and the second data of the number of pieces equal to or larger than a reference value have been received for one or more observation points indicated by the environment data.

5. The information processing apparatus according to claim **1**, wherein the control unit is configured to calculate an average value of respective pieces of data included in the first data for each measurement environment indicated by each piece of data included in the second data in the statistical processing.

6. The information processing apparatus according to claim **1**, wherein the control unit is configured to transmit a result of the statistical processing to another information processing apparatus.

7. An information processing apparatus mounted on a first vehicle, the information processing apparatus comprising a control unit configured to execute:

acquiring one or more pieces of measurement data generated by the first vehicle and one or more pieces of environment data each indicating a measurement environment of each piece of the measurement data;

generating one or more pieces of first dummy data that simulates the measurement data acquired by the first vehicle;

generating one or more pieces of second dummy data that corresponds to the first dummy data and simulates the environment data;

generating first data by adding the first dummy data to one or more pieces of the measurement data;

generating second data by adding the second dummy data to one or more pieces of the environment data;

encrypting each piece of data included in the first data; and

transmitting the first data and the second data to an information processing apparatus mounted on a second vehicle.

8. The information processing apparatus according to claim **7**, wherein the control unit is configured to encrypt each piece of data included in the first data by homomorphic encryption.

9. The information processing apparatus according to claim **7**, wherein:

the environment data is position information of a point at which the measurement data has been acquired; and

the control unit is configured to generate position information of a point on a virtual route as the second dummy data, the virtual route being set as a traveling route of a virtual vehicle simulating the first vehicle.

10. The information processing apparatus according to claim **7**, wherein the control unit is configured to generate position information indicating a predetermined number of positions as the second dummy data, the predetermined number of positions being selected randomly from points on peripheral roads of an observation point indicated by the environment data.

11. The information processing apparatus according to claim **7**, wherein:

the control unit is configured to generate a value of zero as the first dummy data when addition calculation is utilized as statistical processing of each piece of data included in the first data, the statistical processing being performed for each measurement environment indicated by each piece of data included in the second data; and

the control unit is configured to generate a value of one as the first dummy data when multiplication calculation is utilized as the statistical processing.

12. The information processing apparatus according to claim **7**, wherein the control unit is configured to:

set a dummy measurement point at which the first dummy data is assumed to have been measured;

determine whether or not a probability of being able to discriminate an actual measurement point is less than a predetermined threshold when the dummy measurement point and the actual measurement point indicated by the environment data are mixed; and

transmit the first data and the second data to the information processing apparatus mounted on the second vehicle when the control unit determines that the probability of being able to discriminate the actual measurement point is less than the predetermined threshold.

13. The information processing apparatus according to claim **1**, wherein the control unit configured to execute:

receiving a result of statistical processing performed on each piece of data included in the first data that is kept encrypted for each measurement environment indicated by each piece of data included in the second data; and

decrypting the received first data and the received result.

14. An information processing method, comprising:

receiving, from a first vehicle, encrypted first data including one or more pieces of measurement data acquired by the first vehicle and one or more pieces of first dummy data that simulates the measurement data, and second data including one or more pieces of environment data each indicating a measurement environment of each piece of the measurement data and one or more pieces of second dummy data that corresponds to the first dummy data and simulates the environment data; and

performing statistical processing of each piece of data included in the first data for each measurement environment indicated by each piece of data included in the second data.

15. The information processing method according to claim **14**, wherein performing the statistical processing includes executing the statistical processing without decrypting each piece of data included in the first data.

16. The information processing method according to claim **15**, wherein performing the statistical processing includes performing the statistical processing on homomorphically encrypted data.

17. The information processing method according to claim **14**, wherein performing the statistical processing includes performing the statistical processing when the first data and the second data of the number of pieces equal to or larger than a reference value are received for one or more observation points indicated by the environment data.

18. The information processing method according to claim **14**, wherein performing the statistical processing includes calculating an average value of respective pieces of data included in the first data for each measurement environment indicated by each piece of data included in the second data.

19. The information processing method according to claim **14**, further comprising transmitting a result of the statistical processing to another information processing apparatus.

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